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Inventor(s)	Toennies; Heath A.

System to reduce line loss in pressure control hydraulic circuit

Abstract

A hydraulic system including a first cylinder conduit configured to couple to a cylinder, a second cylinder conduit configured to fluidly coupled to the cylinder, and a bypass conduit fluidly coupled both to the first cylinder conduit upstream of the cylinder and to the second cylinder conduit downstream of the cylinder. The bypass conduit is configured to enable intermittent fluid flow of a hydraulic fluid from the first cylinder conduit to the second cylinder conduit while bypassing at least a portion of the cylinder.

Inventors:	Toennies; Heath A. (Eureka, IL)
Applicant:	CNH Industrial America LLC (New Holland, PA)
Family ID:	80930452
Assignee:	CNH Industrial America LLC (New Holland, PA)
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Primary Examiner: Wiehe; Nathaniel E

Assistant Examiner: Collins; Daniel S

Attorney, Agent or Firm: Fletcher Yoder, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority from and the benefit of U.S. Provisional Application Ser. No. 63/164,670, entitled “System to Reduce Line Loss in Pressure Control Hydraulic Circuit,” filed Mar. 23, 2021, which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) The present disclosure relates generally to an agricultural system and, more specifically, to reducing line loss in a pressure control hydraulic circuit of the agricultural system.

(2) Certain agricultural implements include ground engaging tools configured to interact with the soil. For example, a tillage implement may include tillage points and/or disc blades configured to break up the soil for subsequent planting or seeding operations. Tillage implements typically include one or more actuators (e.g., hydraulic cylinders) configured to control a penetration depth of the ground engaging tools into the soil. The actuator(s) may also move the ground engaging tools between a lowered/ground engaging position and a raised/transport portion (e.g., to facilitate

repositioning the tillage implement between successive rows). Furthermore, certain actuators may control the downforce applied by the ground engaging tools to the soil. The actuator(s) are controlled by a hydraulic system including a series of connections (e.g., hoses, lines, etc.) and valves disposed between the actuator(s) and a hydraulic fluid reservoir. Unfortunately, in certain conditions (e.g., cold weather) oil in the lines of the hydraulic system furthest away from the work vehicle (e.g., hydraulically coupled to the implement) remains cold and does not warm up during operation. The high viscosity of the cold oil results in hydraulic losses that hinder achieving the desired performance from the hydraulic system on the agricultural implement.

BRIEF DESCRIPTION

(3) In certain embodiments, a hydraulic system includes a first cylinder conduit configured to couple to a cylinder, a second cylinder conduit configured to fluidly couple to the cylinder, and a bypass conduit fluidly coupled both to the first cylinder conduit upstream of the cylinder and to the second cylinder conduit downstream of the cylinder. The bypass conduit is configured to enable intermittent fluid flow of a hydraulic fluid from the first cylinder conduit to the second cylinder conduit while bypassing at least a portion of the cylinder.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

(2) FIG. 1 is a perspective view of an embodiment of an agricultural system having a work vehicle and an agricultural implement, in accordance with an aspect of the present disclosure;

(3) FIG. 2 is a schematic diagram of an embodiment of a hydraulic system that may be employed to control actuator(s) of the agricultural implement of FIG. 1;

(4) FIG. 3 is a schematic diagram of another embodiment of a hydraulic system that may be employed to control actuator(s) of the agricultural implement of FIG. 1; and

(5) FIG. 4 is a schematic diagram of another embodiment of a hydraulic system that may be employed to control actuator(s) of the agricultural implement of FIG. 1.

DETAILED DESCRIPTION

(6) One or more specific embodiments of the present disclosure will be described below. In an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(7) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Any examples of operating parameters and/or environmental conditions are not exclusive of other parameters/conditions of the disclosed embodiments.

(8) Turning to the drawings, FIG. 1 is a perspective view of an embodiment of an agricultural system 50 that includes a work vehicle 52 and an agricultural implement 54. In the illustrated embodiment, the work vehicle 52 is a tractor. However, in some embodiments, the work vehicle 52

may be an on-road truck, a harvester, and so forth, that may be driven over a field, such as a farming field. As illustrated, the work vehicle **52** includes a cab **56** mounted on a chassis **58**. The chassis **58** may support components, such as a motor, a hydraulic system (e.g., a pump, valves, a reservoir), an electrical system (e.g., a control system), a cooling system (e.g., an engine coolant system, a heating, ventilation, and/or air conditioning system), and the like, to facilitate operation of the work vehicle **52**. Additionally, the work vehicle **52** includes tracks **60** (e.g., wheels) that operate to move the work vehicle **52**. For example, the front and/or the rear tracks **60** may rotate in a first rotational direction **62** (e.g., a forward rotational direction) about a lateral axis **63** to drive the work vehicle **52** in a first direction **64** (e.g., a forward direction), and the front and/or rear tracks **60** may rotate in a second rotational direction **66** (e.g., reverse rotational direction) about the lateral axis **63**, opposite the first rotational direction **62**, to drive the work vehicle **52** in a second direction **68** (e.g., backward direction), opposite the first direction **64**. The tracks **60** (e.g., the front tracks and/or the rear tracks) may also be steered to turn the work vehicle **52**. In additional or alternative embodiments, a portion (e.g., a rear portion) of the chassis may rotate relative to a remaining portion (e.g., a front portion) of the chassis to steer the work vehicle.

(9) The cab **56** is configured to house an operator of the work vehicle **52** during operation of the agricultural system **50**. The cab **56** may provide access to various controls of the work vehicle **52**. For example, the cab **56** may include a user interface to enable the operator to control the operation of certain systems of the work vehicle **52**. In some embodiments, the cab **56** may include a component, such as a steering wheel, to enable the operator to steer the tracks **60** to turn the work vehicle **52**. In addition, the cab may include other and/or additional types of user interfaces (e.g., a touch screen, a hand controller, a push button, a track pad) configured to receive user input or feedback for controlling various operations and systems of the work vehicle.

(10) Moreover, the chassis **58** is coupled to the agricultural implement **54** to enable the work vehicle **52** to tow the agricultural implement **54**. For example, the chassis **58** may be coupled to a hitch **70** of the agricultural implement **54** (e.g., via a corresponding hitch of the work vehicle). In addition, the agricultural implement **54** includes main wheels **72** that enable the agricultural implement **54** to move, such as over the field through which the work vehicle **52** is navigating. Thus, movement of the work vehicle **52** drives movement of the agricultural implement **54**. For example, movement of the work vehicle **52** in the first direction **64** drives the agricultural implement **54** to move in the first direction **64**, and movement of the work vehicle **52** in the second direction **68** drives the agricultural implement **54** to move in the second direction **68**. In certain embodiments, the agricultural implement **54** may also be steerable. By way of example, the main wheels **72** may be turned to steer the agricultural implement **54**.

(11) In some embodiments, the agricultural implement **54** may be a tilling implement (e.g., vertical tilling implement) configured to break up soil within the field during operation of the agricultural system **50**. The agricultural implement **54** includes a frame **74** to which the main wheels **72** are coupled. As illustrated in FIG. 1, the main wheels **72** are located between a first end **76** (e.g., front end) of the agricultural implement **54** and a second end **78** (e.g., rear end) of the agricultural implement **54**. The agricultural implement **54** includes gauge wheels **80** that are coupled to the frame **74**, such as at the first end **76**. The gauge wheels **80** may be used to reduce an amount of lateral and/or vertical movement of the agricultural implement **54** while the agricultural system **50** is in operation. For example, the gauge wheels **80** may engage the soil surface while the work vehicle **52** tows the agricultural implement **54**, such that movement of the agricultural implement **54** along the lateral axis **63** and/or movement of the agricultural implement **54** along a vertical axis **84** is reduced.

(12) The agricultural implement **54** also includes disc blades **86** that are coupled to the frame **74**. In the illustrated embodiments, the disc blades **86** of the agricultural implement **54** are arranged in rows, including a first blade row **88**, a second blade row **90**, a third blade row **92**, and a fourth blade row **94**. The first blade row **88** and the second blade row **90** may each be positioned at the

first end **76** of the agricultural implement **54**, and the third blade row **92** and the fourth blade row **94** may each be positioned at the second end **78** of the agricultural implement **54**. In some embodiments, the first blade row **88** may be positioned adjacent to and at an angle with respect to the second blade row **90**, and the third blade row **92** may be positioned adjacent to and at an angle with respect to the fourth blade row **94**. Thus, the blade rows **88**, **90**, **92**, **94** may form an x-shaped configuration on the agricultural implement **54**. In additional or alternative embodiments, the disc blades may be arranged in a different configuration on the agricultural implement (e.g., a k-shaped configuration, a diamond configuration, a parallel configuration), and/or the agricultural implement may include a different number of rows of disc blades.

(13) During operation of the agricultural system **50**, the disc blades **86** may engage soil of the field. For instance, the main wheels **72** may be positioned to set the position of the frame **74** at a target height above the soil surface. By way of example, the main wheels **72** may move (e.g., translate, rotate) away from the frame **74** to drive the frame **74** away from the soil surface, and the main wheels **72** may move toward the frame **74** to drive the frame **74** toward the soil surface. As the agricultural implement **54** is towed by the work vehicle **52**, the disc blades **86** may rotate while engaged with the soil to till the soil. Each disc blade **86** may, for example, be non-translatably coupled to the frame **74**, such that movement of the main wheels **72** relative to the frame **74** changes the position of the disc blades **86** relative to the soil surface (e.g., to engage or disengage the disc blades **86** from the soil). In some embodiments, the disc blades **86** may be concave or may have certain surface features (e.g., flutes) that facilitate tilling of the soil. In additional or alternative embodiments, the agricultural implement **54** may include other suitable type(s) of ground engaging tool(s), such as tillage points, tines, and so forth.

(14) Although the position of the disc blades **86** relative to the soil surface is adjusted by controlling the position of the main wheels in the illustrated embodiment, in additional or alternative embodiments, the position of the disc blades may be adjusted in other manners. For example, the hitch may be raised and/or lowered relative to the soil surface (e.g., via adjustment of the hitch of the work vehicle) to adjust the position and/or a pitch of the agricultural implement, thereby adjusting the position of the disc blades relative to the soil surface. In further embodiments, the disc blades may be directly adjustable relative to the implement frame. For example, groups of disc blades may be adjustable via a subframe of the agricultural implement.

(15) The agricultural implement **54** may include basket assemblies **96** (e.g., crumbler baskets), which may be disposed at the second end **78** of the agricultural implement **54**. The basket assemblies **96** are configured to engage the soil surface during operation of the agricultural system **50**. For example, as the work vehicle **52** tows the agricultural implement **54** in the first direction **64**, each basket assembly **96** may provide a downward force on the soil and rotate to level the soil tilled by the disc blades **86** and/or to cut crop residue in the field. Although the agricultural implement **54** has three basket assemblies **96** in the illustrated embodiment, additional or alternative embodiments of the agricultural implement may have any suitable number of basket assemblies and/or any other suitable component(s) (e.g., tines) configured to level the soil during operation of the agricultural system. In some embodiments, the basket assemblies **96** may also stabilize the agricultural implement **54** during operation of the agricultural system **50**. For example, the basket assemblies **96** may dampen vertical movement of the agricultural implement **54** by providing a downward pressure. The agricultural implement may have any suitable alternate configuration, such as having no gauge wheels, no main wheels, no baskets, any other suitable configuration, or any combination thereof. The agricultural implement may also be any other suitable type of agricultural implement, such as a planting implement, a seeding implement, and so forth.

(16) In the illustrated embodiment, the agricultural implement **54** includes downforce actuators **122** (e.g., linear actuators) configured to control a downforce applied by the basket assemblies **96** to the soil. The downforce actuator **122** extends from the frame **74** to a frame **98**, which is pivotally

coupled to the frame **74**. As discussed in detail below, a hydraulic system is fluidly coupled to the actuators **122** and configured to control the actuators **122** during operation of the agricultural implement **54**, thereby controlling the downforce applied by basket assemblies **96** to the soil. The actuators **122** can also move the basket assemblies **96** between lowered and raised positions. In certain embodiments, the hydraulic system includes a supply conduit configured to provide fluid (e.g., hydraulic fluid) to the actuator **122** and a return conduit configured to return the fluid to a reservoir.

(17) In addition, while the hydraulic system described herein is used to control the downforce applied by the basket assemblies **96**, in other embodiments, the hydraulic system may be utilized to control the force applied by other ground engaging tool(s) of the agricultural implement, the position of other ground engaging tool(s) of the agricultural implement, or a combination thereof. Furthermore, in certain embodiments, the hydraulic system may be utilized to control the force applied by/position of at least one ground engaging tool of another suitable implement (e.g., a planting implement, a seeding implement, a harvesting implement, etc.).

(18) FIG. **2** is a schematic diagram of an embodiment of the hydraulic system **112** that may be employed to control actuator(s) of the agricultural implement of FIG. **1**. In the illustrated embodiment, the implement hydraulic system **112** is configured to control the downforce applied by the basket assemblies **96** to the soil. However, in other embodiments, the hydraulic system **112** may control the position of the basket assemblies **96** relative to the soil, or the hydraulic system **112** may control the force applied by/position of another suitable ground engaging tool.

(19) In the illustrated embodiment, the hydraulic system **112** includes a supply conduit **142** configured to receive fluid (e.g., hydraulic fluid) from a reservoir **140**, disposed on the work vehicle. The hydraulic system **12** also includes a return conduit **144** configured to return the fluid to the reservoir **140**, disposed on the work vehicle. In addition, the hydraulic system **112** includes a pump **146** (e.g., hydraulic pump) disposed on the work vehicle, and the reservoir **140** (e.g., including a hydraulic fluid tank) is disposed on the work vehicle. The pump **146** is fluidly coupled to the reservoir **140** and configured to circulate hydraulic fluid through the hydraulic system **112**.

(20) Furthermore, the hydraulic system **112** includes a cylinder control valve **148** fluidly coupled to the supply conduit **142** and the return conduit **144**. The cylinder control valve **148** is also fluidly coupled to a first cylinder conduit **150** and a second cylinder conduit **152** that are disposed on the implement. In the illustrated embodiment, the first cylinder conduit **150** is fluidly coupled to a pressure regulator **174** (e.g., pressure reducing and relieving valve) and a cylinder cap end **154** (e.g., first end) of the actuator **122** (e.g., double-acting cylinder), and the second cylinder conduit **152** is fluidly coupled to a rod end **156** (e.g., second end) of each actuator **122** (e.g., double-acting cylinder). The actuators **122** may be located near or at a rear portion of the agricultural implement (e.g., furthest away from the hitch). The pressure regulator **174** may be disposed adjacent a front portion of the agricultural implement (e.g., nearest the hitch). Thus, the hydraulic lines (e.g., first cylinder conduit **150** and the second cylinder conduit **152**) may be a lengthy distance between the pressure regulator **174** and the actuators **122** making the hydraulic fluid in the lines subject to cold weather conditions. The pressure regulator **174** regulates the downforce applied by the basket assemblies via the actuators **122** when valve **168** connects fluid conduit **150** to the outlet of pump **146** and connects fluid conduit **152** to the return conduit **144**. The pressure regulator **174**, which is discussed in more detail below, enables fluid to flow through the first cylinder conduit **150** and blocks fluid flow from the first cylinder conduit **150** to the second cylinder conduit **152** while the fluid pressure within the first cylinder conduit **150** on the cylinder side of the pressure regulator **174** (e.g., within the cylinder cap end **154**) is less than or equal to a threshold pressure. Providing fluid to the cylinder cap end **154** of each actuator **122** drives a respective piston rod **58** to extend and providing fluid to the rod end **156** of each actuator **122** drives the respective piston rod **158** to retract. In the illustrated embodiment, the extension force of the piston rod **158** increases the downforce applied by the basket assemblies (e.g., driving the basket assemblies into the ground),

and is regulated by pressure control valve **174**. Retraction of the piston rod **158** raises the basket off the ground when it is not in use. However, in other embodiments, extension of the piston rod may raise the baskets off the ground, and retraction of the piston rod may increase the downforce. In such embodiments, the first cylinder conduit **150** may be fluidly coupled to the pressure regulator **174** and to the rod end (e.g., first end) of the actuator, and the second cylinder conduit **152** may be coupled to the cap end (e.g., second end) of the actuator.

(21) As depicted, multiple actuators **122** are utilized to control the downforce applied to the barrel assemblies. The number of actuators **122** may vary (e.g., 1, 2, 3, 4, or more). The actuators **122** are fluidly coupled in to one another in a parallel arrangement. While the illustrated actuators **122** are configured to control the downforce applied by the basket assemblies, in other embodiments, the actuator may be configured to control the downforce applied by other suitable ground engaging tool(s) (e.g., disc blade(s), one or more tillage point assemblies, etc.).

(22) In the illustrated embodiment, the cylinder control valve **148** is a proportional three position/four way valve. The first position **160** of the cylinder control valve **148** is configured to block fluid flow between the supply conduit **142** and the first and second cylinder conduits and between the return conduit **144** and the first and second cylinder conduits, thereby blocking fluid flow between the supply conduit **142** and both ends of the actuator **122** and between the return conduit **144** and both ends of the actuator **122**. The second position **162** of the cylinder control valve **148** is configured to facilitate fluid flow between the supply conduit **142** and the rod end **156** of the actuator **122** (e.g., via the second cylinder conduit **152**) and between the return conduit **144** and the cylinder cap end **154** of the actuator **122** (e.g., via the first cylinder conduit **150**) to drive the actuator **122** to raise the basket assembly away from the soil. The third position **164** of the cylinder control valve **148** is configured to facilitate fluid flow between the supply conduit **142** and the cylinder cap end **154** of the actuator **122** and between the return conduit **144** and the rod end **156** of the actuator **122** to drive the actuator **122** to increase the downforce applied by the basket assembly to the soil surface. In the illustrated embodiment, the cylinder control valve **148** is a proportional control valve configured to control the fluid flow rate through the valve (e.g., based on the position of the valve relative to the first position). However, in other embodiments, the cylinder control valve may be another suitable type of valve configured to control fluid flow between the supply and return conduits and the actuator.

(23) In the illustrated embodiment, the cylinder control valve **148** includes a retract actuator **166** configured to drive the cylinder control valve **148** to the second position **162**. And, the cylinder control valve **148** includes an extend actuator **168** configured to drive the cylinder control valve **148** to the third position **164**. In the illustrated embodiment, the retract actuator **166** and the extend actuator **168** are electronic actuators (e.g., solenoids) configured to move the cylinder control valve **148** in response to receiving an electrical signal. In addition, the cylinder control valve **148** includes biasing elements **170** (e.g., springs) configured to urge the cylinder control valve **148** toward the first position **160**. Accordingly, applying an electrical signal to the retract actuator **166** drives the cylinder control valve **148** to the second position **162**, thereby causing the actuator **122** to raise the basket away from the soil. Furthermore, applying an electrical signal to the extend actuator **168** drives the cylinder control valve **148** to the third position **164**, thereby causing the actuator **122** to increase the downforce applied by the basket assemblies to the soil. Furthermore, if no electrical signal is applied to either actuator, the biasing elements **170** drive the cylinder control valve **148** to the first position **160**, thereby blocking fluid flow between the supply and return conduits and the actuator **122**.

(24) In the illustrated embodiment, the hydraulic system **112** includes bypass conduits **176** extending between the first cylinder conduit **150** and the second fluid conduit **152** that enable fluid flow (e.g., intermittent fluid flow) of the hydraulic fluid (e.g., oil) to bypass the actuators **122** during operation of the hydraulic system **112**. Even though multiple bypass conduits **176** are illustrated, the number of bypass conduits **76** may vary (e.g., 1, 2, 3, 4, 5, or more). As depicted, the

bypass conduits **176** are arranged between the actuators **122** in alternating pattern, the arrangement of the bypass conduits **176** may vary. For example, one or more bypass conduits **176** may be disposed on the outside of the actuators **122** (e.g., a pair of bypass conduits **176** may flank the actuators **122**).

(25) One or more flow features may be disposed along the bypass conduits **176** to promote flow of the hydraulic fluid from the first cylinder conduit **150** to the second cylinder conduit **152** via the bypass conduits **176** while bypassing the actuators **122**. In the illustrated embodiment, an orifice **178** (e.g., bleed orifice) is disposed along each bypass conduit **176** to promote flow of the hydraulic fluid from the first cylinder conduit **150** to the second cylinder conduit **152** (e.g., from the pressure controlled end (via the pressure regulator **174**) of the cylinder circuit to the other side of the cylinder circuit). The bleed orifice **178** also enables the purge of air and potentially other contaminants that may build up in portions of the hydraulic lines in the absence of continuously circulating hydraulic flow. In certain embodiments, a check valve **180** may be disposed along each bypass conduit **176**. As depicted, each check valve **180** is disposed upstream (e.g., relative to a flow direction from the first cylinder conduit **150** the second cylinder conduit **152** via the bypass conduit **176**). The check valve **180** enables hydraulic fluid to flow along the bypass conduit **176** (from the first cylinder conduit **150** to the second cylinder conduit **152**) while keeping hydraulic fluid from flowing back in the opposite direction towards the first cylinder conduit **150**. Blocking hydraulic fluid flow back in the opposite direction enables a ground engaging tool (e.g., basket assembly) to be held in a raised position when not in use. Thus, enabling the load holding function to be preserved while also enabling improved downpressure control. In certain embodiments, where the load holding function is not needed, the check valve **180** may not be present. In certain embodiments, as illustrated, a filter **182** (e.g., screen filter) is disposed along each bypass conduit **176** upstream of both the bypass conduit **176** and the check valve **180**. The filter **182** keeps contaminants or particulates from entering into the orifice **178** and blocking or hindering fluid flow through the orifice. In certain embodiments, the filter **182** may be a different type of filter from the screen filter as long the as the filter keeps the contaminants from entering the orifice **178**. As described in greater detail below, flow features other than the bleed orifice **178** may be utilized to enable hydraulic fluid flow from the first cylinder conduit **150** to the second cylinder fluid conduit **152** via the bypass conduits **176**.

(26) In the absence of the bypass conduits **76**, in cold weather, the hydraulic fluid in the lines (e.g., portions of the first and second cylinder conduits **150**, **152**) becomes cold and increases in viscosity due to lack of circulation. Even during operation of the implement, the hydraulic fluid never warms up (e.g., due to cold, high viscosity hydraulic fluid remaining in the lines or due to the hydraulic fluid in the lines cooling due to heat transfer to the cold ambient environment). Instead, the cold, high viscosity hydraulic fluid remains in the first cylinder conduit **150** near the actuators **122** or moves back and forth between the pressure regulator **174** and the actuators **122**. This would result in excessive line losses (e.g., hydraulic losses) causing pressure errors (and pressure spikes) in the pressure control circuit for the actuators **122**. For example, pressure at the actuators may be significantly higher or lower than the nominal pressure setting at the valve due to the excessive line losses caused by cylinder motion with cold, high viscosity hydraulic fluid.

(27) The bypass conduits **176** and associated flow features address the problems that cold weather may cause to the hydraulic system **112**. The first cylinder conduit **150**, the bypass conduits **176**, and the second cylinder conduit **152** (along with the supply conduit **142** and the return conduit **144**) form a circulation conduit **184** for a circulating flow of the hydraulic fluid (e.g., of warm hydraulic fluid from the work vehicle to the actuators **122** and back) as indicated by the arrows **186**.

Continuous circulating flow of the hydraulic fluid along the circulation conduit **184**, while the implement is operating, enables maintenance of warm, low viscosity hydraulic fluid in the lines for improved performance (e.g., due to reduced hydraulic fluid viscosity, reduced line losses, reduced variations in pressure at the actuators **122**, etc.), especially in hydraulic systems with long hydraulic

lines between the pressure control system and the actuators.

(28) One or more components of the hydraulic system **112** (e.g., pump **146**, cylinder control valve **148**, pressure regulator **174**, certain flow features disposed along the bypass conduits **176**, etc.) may be controlled by a control system **188**. In particular, the control system **188** outputs control signal(s) to various components, such as to control power flow to the components to move and operate the components. The control system **188** includes a memory **190** and processing circuitry **192**. The memory **90** may include volatile memory, such as random-access memory (RAM), and/or non-volatile memory, such as read-only memory (ROM), optical drives, hard disc drives, solid-state drives, or any other non-transitory computer-readable medium that includes instructions executable by the processing circuitry **192**. The processing circuitry **192** may include one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), one or more general purpose processors, or any combination thereof, configured to execute the instructions stored in the memory **190** to operate the agricultural system (e.g., agricultural implement). For example, the instructions stored in the memory **190** cause the processing circuitry **192** to output a control signal to components of the hydraulic system **112** and direct fluid through the hydraulic system **112** to operate the agricultural system.

(29) FIG. **3** is a schematic diagram of another embodiment of a hydraulic system **194** that may be employed to control actuator(s) of the agricultural implement of FIG. **1**. The hydraulic system **194** operates similar to the hydraulic system **112** in FIG. **2**. However, the hydraulic system **194** includes a different type of flow feature. As illustrated, the hydraulic system **194** includes a valve **196** (illustrated by a generic valve) disposed along each bypass conduit **176** to regulate hydraulic flow (e.g., intermittent flow) from the first cylinder conduit **150** to the second cylinder conduit **152** while bypassing the actuators **122**. In certain embodiments, the valve **196** may be electronic actuated directional valve (e.g., controlled by control signals from the control system **188**). In other embodiments, the valve **196** may be pressure relief valve (e.g., controlled by control signals from the control system **188**).

(30) In certain embodiments, as depicted in FIG. **3**, the hydraulic system **194** includes a temperature monitoring system **198** (e.g., thermostat device). The thermostat system **198** includes one or more sensors **200** (e.g., temperature transducers) to monitor a temperature of the hydraulic fluid. The sensors **200** may be disposed along the first cylinder conduit **150**, the bypass conduits **176**, and/or the second cylinder conduit **152**. The sensors **200** provide feedback to the temperature monitoring system. The temperature monitoring system **198** may be communicatively coupled to the control system **188**. Based on the feedback from the sensors **200**, if the temperature of the hydraulic fluid reaches a certain threshold temperature (and, thus, the hydraulic fluid is warm enough to not adversely affect hydraulic operations), the flow features (e.g., valves **196**) may be closed to stop the intermittent flow of hydraulic fluid from the first cylinder conduit **150** to the second cylinder conduit **152** via the bypass conduits **176**. If the temperature of the hydraulic fluid has not reached a certain threshold temperature (e.g., is lower than the threshold temperature), the flow features (e.g., valves **196**) remain open to enable the intermittent flow of the hydraulic fluid across the bypass conduits **176**. Although the temperature monitoring system **198** is utilized in conjunction with the flow features in FIG. **3**, the temperature monitoring system **198** may be utilized with other flow features.

(31) FIG. **4** is a schematic diagram of another embodiment of a hydraulic system **202** that may be employed to control actuator(s) of the agricultural implement of FIG. **1**. Unlike the hydraulic systems **112**, **194** in FIGS. **2** and **3**, the hydraulic system **202** does not utilize separate bypass conduits with flow features that promote intermittent flow of the hydraulic fluid from the first cylinder conduit **150** to the second cylinder conduit **152** bypassing the actuators. Instead, the hydraulic system **202** utilizes rephasing cylinders **204** as the actuators. Each rephasing cylinder **204** includes the bypass conduit **176** to enable hydraulic fluid to bypass the cylinder piston seal (and thus the piston) when the piston is either in a fully extended or retracted position. The bypass

conduit **176** may include a passageway within the wall or on the inside of the wall of the cylinder **204**. The bypass conduits **176** enable the hydraulic fluid to pass from the first cylinder conduit **150** to the second cylinder conduit **152** while bypassing a portion of the cylinder **204** (e.g., the piston). The bleeding of the hydraulic fluid promotes circulation of the hydraulic fluid along the circulation conduit **184**.

(32) While only certain features have been illustrated and described herein, many modifications and changes will occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure.

(33) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

Claims

1. A hydraulic system comprising: a first cylinder conduit configured to fluidly couple to a cylinder; a second cylinder conduit configured to fluidly coupled to the cylinder; and a bypass conduit fluidly coupled both to the first cylinder conduit at a first end of the cylinder and to the second cylinder conduit at a second end of the cylinder opposite the first end, wherein the bypass conduit is configured to enable intermittent fluid flow of a hydraulic fluid from the first cylinder conduit to the second cylinder conduit while bypassing at least a portion of the cylinder, wherein the cylinder comprises a rephasing cylinder having a piston, wherein the rephasing cylinder comprises the bypass conduit, and the bypass conduit is configured to enable the intermittent fluid flow of the hydraulic fluid from the first cylinder conduit to the second cylinder conduit while bypassing the piston of the rephasing cylinder.
2. The hydraulic system of claim 1, wherein the first cylinder conduit, the bypass conduit, and the second cylinder conduit form a portion of a circulation conduit that enables circulating flow of the hydraulic fluid along the circulation conduit that causes the hydraulic fluid to be at a viscosity that minimizes hydraulic losses along the circulation conduit.
3. A hydraulic system, comprising: a first cylinder conduit; a second cylinder conduit; a cylinder fluidly coupled to the first cylinder conduit and the second cylinder conduit; a bypass conduit fluidly coupled both to the first cylinder conduit at a first end of the cylinder and to the second cylinder conduit at a second end of the cylinder opposite the first end, wherein the bypass conduit is configured to enable intermittent fluid flow of a hydraulic fluid from the first cylinder conduit to the second cylinder conduit while bypassing the cylinder, and wherein the first cylinder conduit, the bypass conduit, and the second cylinder conduit form a circulation conduit configured to enable circulating flow of the hydraulic fluid along the circulation conduit; a valve disposed along the bypass conduit, wherein the valve is configured to enable intermittent fluid flow of the hydraulic fluid from the first cylinder conduit to the second cylinder conduit while bypassing the cylinder; a plurality of temperature sensors disposed along the first cylinder conduit, the second cylinder conduit, and the bypass conduit; and a temperature monitoring system configured to monitor a temperature of the hydraulic fluid based on feedback from the plurality of temperature sensors and to cause blocking of the intermittent flow of the hydraulic fluid from the first cylinder conduit to the second cylinder conduit along the bypass conduit when the temperature reaches a threshold temperature by closing the valve.
4. A hydraulic system for an agricultural implement having a ground engaging tool, comprising: a

first cylinder conduit; a second cylinder conduit; a cylinder fluidly coupled to the first cylinder conduit and the second cylinder conduit; a bypass conduit fluidly coupled both to the first cylinder conduit at a first end of the cylinder and to the second cylinder conduit at a second end of the cylinder opposite the first end; and a bleed orifice disposed along the bypass conduit, wherein the bleed orifice is configured to enable intermittent fluid flow of a hydraulic fluid from the first cylinder conduit to the second cylinder conduit while bypassing the cylinder while the ground engaging tool is being actuated by the hydraulic system.

5. The hydraulic system of claim 4, comprising a check valve disposed along the bypass conduit, wherein the check valve is configured to block fluid flow back to the first fluid conduit along the bypass conduit.

6. The hydraulic system of claim 5, wherein the check valve is disposed upstream of the bleed orifice along the bypass conduit.

7. The hydraulic system of claim 5, comprising a screen filter disposed along the bypass conduit, wherein the screen filter is disposed upstream of the check valve and is configured to keep contaminants from entering the check valve and blocking fluid flow through the check valve.

8. The hydraulic system of claim 4, wherein the first cylinder conduit, the bypass conduit, and the second cylinder conduit form a circulation conduit configured to enable circulating flow of the hydraulic fluid along the circulation conduit.
